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PETERS

SL 1407 FLZ-S



- good protection against corrosion
- very good dielectric properties
- fluorescent
- good yellowing resistance
- can be soldered through for repair purposes or removed with the help of thinner V 1407 FLZ

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Contact :

EMS Technologies

#G3 Reliance Residency

500029 Hyderabad

Email: info@emstechnologies.in

Ph: +91-40-66567767

Cell: +91-9133707031

www.peters-india.com

Q1. What is the use of nanofarad (nF) or picofarad (pF) non-electrolytic capacitors in a circuit? What is the difference between non-electrolytic and electrolytic capacitors in alternating-current (AC) and direct-current (DC) power circuits?

Amitava Sen
Kolkata

A1. Capacitors are common components of electronic circuits. Unlike resistor, the behaviour of a capacitor is frequency-dependent. Capacitive reactance (the opposition to alternating current due to capacitance) of a capacitor is high at low frequencies. The same capacitor offers a low reactance at high frequencies.

Capacitors can be broadly categorized into polarised and non-polarised. Further, these are classified on the basis of dielectric material used. Other classification is fixed or variable type of capacitors.

Electrolytic capacitors fall in polarised category, meaning these have positive and negative leads, which is marked on the capacitor's body. The polarity of the capacitor is to be maintained while connecting it in the circuit. The positive lead or pin has to be connected to the point with a higher positive voltage than the negative lead. Electrolytic capacitors are available in large μF values (1 μF and above).

Another parameter is the working voltage of the capacitor, which is

also marked on the capacitor's body. The capacitor will explode if the rated working voltage is exceeded. To be on the safe side, select a capacitor with a working voltage of at least 10-15 per cent higher than the voltage in the circuit.

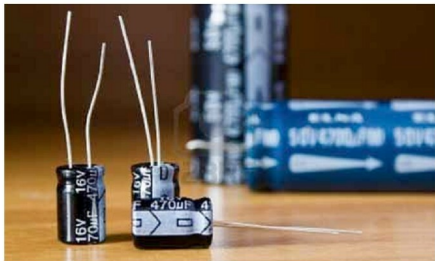
Electrolytic capacitors have a tendency to drift from their values over time, so these are not suitable for use in high-frequency circuits. Electrolytic capacitors are used for signal coupling as well as decoupling.

Ceramic, polyester and mica capacitors fall in non-polarised category. These are available in smaller values in micro-, nano- and pico-Farads. Small-value capacitors are used in circuits for different purposes. These are common components of filtering, tuning, oscillator and timing circuits. Applications in power circuits are as under:

Smoothing out voltage. Power supplies convert AC into DC. After rectification, pulsating DC voltage is received. A high-value capacitor helps in smoothing out the voltage so that it is at a constant level but with a ripple.

Power factor correction. Inductive loads such as induction motors introduce a lagging power factor. A large lagging power factor is not acceptable to the power supplying agencies. Large capacitors are used for power factor correction.

Motor starter capacitor. It is used



Electrolytic capacitors